Widest-Path vs. Shortest-Path Routing Under CSMA/Bianchi WiFi Interference

ncsim Experiment Report

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Abstract

We compare **widest-path** and **shortest-path** multi-hop routing strategies for DAG-scheduled task graphs on wireless mesh networks using the ncsim discrete-event simulator with its expected-rate CSMA/Bianchi interference model and HEFT scheduling. On **grid mesh topologies** (9 scenarios: 2×2, 3×3, 4×4 grids $\times$ 5, 10, 20 tasks), shortest-path wins 7 of 9 cases (2 ties), with widest-path slowdowns up to 4.4× on larger networks. The root cause is *flow concentration*: widest-path funnels transfers onto high-bandwidth links, creating CSMA contention. However, on a **bottleneck relay network** where a slow direct link (1.075 MB/s) competes with a high-bandwidth relay chain (8.6 MB/s per hop), widest-path wins all 3 scenarios with speedups up to 2.5×. The results demonstrate that optimal routing depends on network structure: shortest-path is preferred for dense meshes, while widest-path excels when relay paths offer substantially higher bandwidth than direct links.

1 Introduction

When scheduling DAG-structured computations across a wireless mesh network, data must be transferred between nodes after each task completes. The *routing strategy* determines which multi-hop path each transfer takes. Two natural choices are:

- **Widest path**: maximize the minimum link bandwidth along the route (Dijkstra-style, maximizing bottleneck capacity).
- **Shortest path**: minimize the number of hops (BFS-style, minimizing hop count).

In isolation, widest-path routing selects higher-throughput links. However, in a wireless network where all links within carrier-sensing range contend for airtime via CSMA, concentrating flows on fewer "best" links can create severe contention. Conversely, when the shortest path traverses a slow bottleneck link while an alternative relay chain offers much higher bandwidth, widest-path routing can dramatically outperform shortest-path. This report quantifies both effects using ncsim's expected-rate CSMA/Bianchi interference model across grid mesh and bottleneck relay topologies.

2 Experimental Setup

2.1 Network Topologies

We use square grid mesh networks with 40 m spacing between adjacent nodes. Links include all grid edges (horizontal, vertical) and checkerboard-pattern diagonals, all bidirectional. The WiFi PHY rates are computed from the 802.11ax MCS table based on SNR at each link distance: 40 m grid links and ≈56.6 m diagonal links receive different PHY rates, giving widest-path and shortest-path different preferred routes.

Table 1: Network configurations.

Label	Grid	Nodes	Bidirectional Links
Small	2×2	4	12
Medium	3×3	9	32
Large	4×4	16	68

Nodes have heterogeneous compute capacities (80–300 compute_units/s) so that HEFT distributes tasks across multiple nodes, creating inter-node data transfers.

2.2 DAG Structures

Three DAG sizes capture different communication patterns:

Table 2: DAG configurations (all tasks: compute_cost=500, all edges: data_size=10 MB).

Label	Tasks	Shape
Small	5	Fork-join: 1 source → 3 parallel → 1 sink
Medium	10	Diamond: source → 4 → 4 → sink
Large	20	Multi-level: 5-stage pipeline with branching

2.3 Simulation Parameters

All simulations use:

- **Scheduler**: HEFT (Heterogeneous Earliest Finish Time)
- **Interference**: CSMA/Bianchi (802.11ax, 5 GHz, 20 dBm TX, path loss exponent 3.0)
- **Carrier sensing range**: 71.2 m (all links within each grid are in the conflict graph)
- **Seed**: 42 (deterministic)

The CSMA/Bianchi model combines two interference mechanisms: (1) *Bianchi contention* for links sharing a conflict-graph edge (symmetric MAC-layer time-sharing), and (2) *SINR degradation* for hidden terminals outside the conflict graph. In our grid topologies, nearly all links are within carrier-sensing range, so Bianchi contention dominates.

3 Aggregate Results

The dominant trend is clear: shortest-path routing wins in all non-trivial grid scenarios. The two ties occur with the small DAG on small networks, where both strategies use the same single-hop path. The widest-path overhead ranges from 23% (4×4 medium DAG) to 338% (4×4 small DAG), with the largest absolute slowdown on the 4×4 grid where widest-path routes transfers through long perimeter corridors.

Table 3: Makespan (seconds) for all 18 grid simulations. Bold indicates the winner in each cell. Percentage shows widest-path overhead relative to shortest-path.

		Network Size		
DAG	Routing	2×2 (4)	3×3 (9)	4×4 (16)
Small (5)	Widest	**11.43**	**12.37**	39.15
	Shortest	**11.43**	**12.37**	**8.93**
	Diff	*tie*	*tie*	*+338%*
Medium (10)	Widest	30.02	42.76	124.86
	Shortest	**20.52**	**32.33**	**101.62**
	Diff	*+46%*	*+32%*	*+23%*
Large (20)	Widest	51.65	70.19	221.73
	Shortest	**36.66**	**51.36**	**160.62**
	Diff	*+41%*	*+37%*	*+38%*

Table 4: Winner summary for grid topologies: shortest-path wins in 7 of 9 cases.

	2×2	3×3	4×4
Small DAG	Tie	Tie	Shortest
Medium DAG	Shortest	Shortest	Shortest
Large DAG	Shortest	Shortest	Shortest

4 Per-Scenario Analysis

Each figure below shows the widest-path (left) and shortest-path (right) routing side by side. The top row displays the network topology with links colored and thickened by flow count (number of data transfers using that link). The middle row shows a Gantt-style timeline of task executions (blue) and data transfers (orange). The bottom row shows per-directed-link flow counts.

4.1 2×2 Network $\times$ Small DAG (5 tasks)

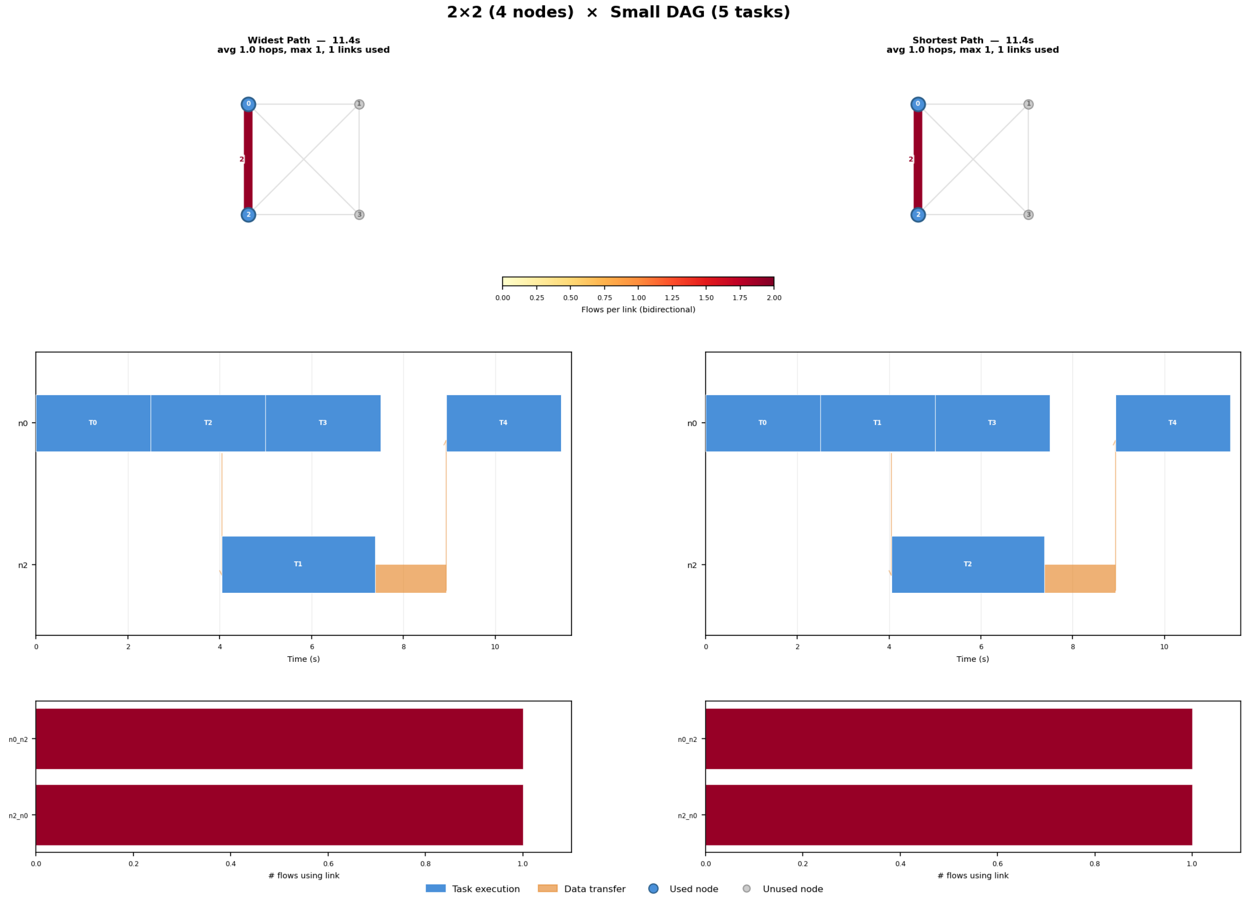

Figure 1: $2 \times 2 \times$ Small DAG. Both routing strategies produce identical results.

With only 4 nodes and a simple fork-join DAG, HEFT places 4 of 5 tasks on the fastest node (n0, capacity=200) and sends one task to n2. Both strategies use the same single-hop direct link (n0↔n2) for both transfers. There is no routing decision to make: the network is too small for path diversity.

4.2 2×2 Network $\times$ Medium DAG (10 tasks)

With 10 tasks across 3 nodes, widest path introduces 2-hop routes through diagonal links, increasing total link occupancy and CSMA contention. Under the expected-rate SINR model, the additional hidden-terminal interference from multi-hop routes outweighs any bandwidth advantage. Shortest path keeps all transfers on direct 1-hop links, minimizing the contention footprint and finishing 46% faster.

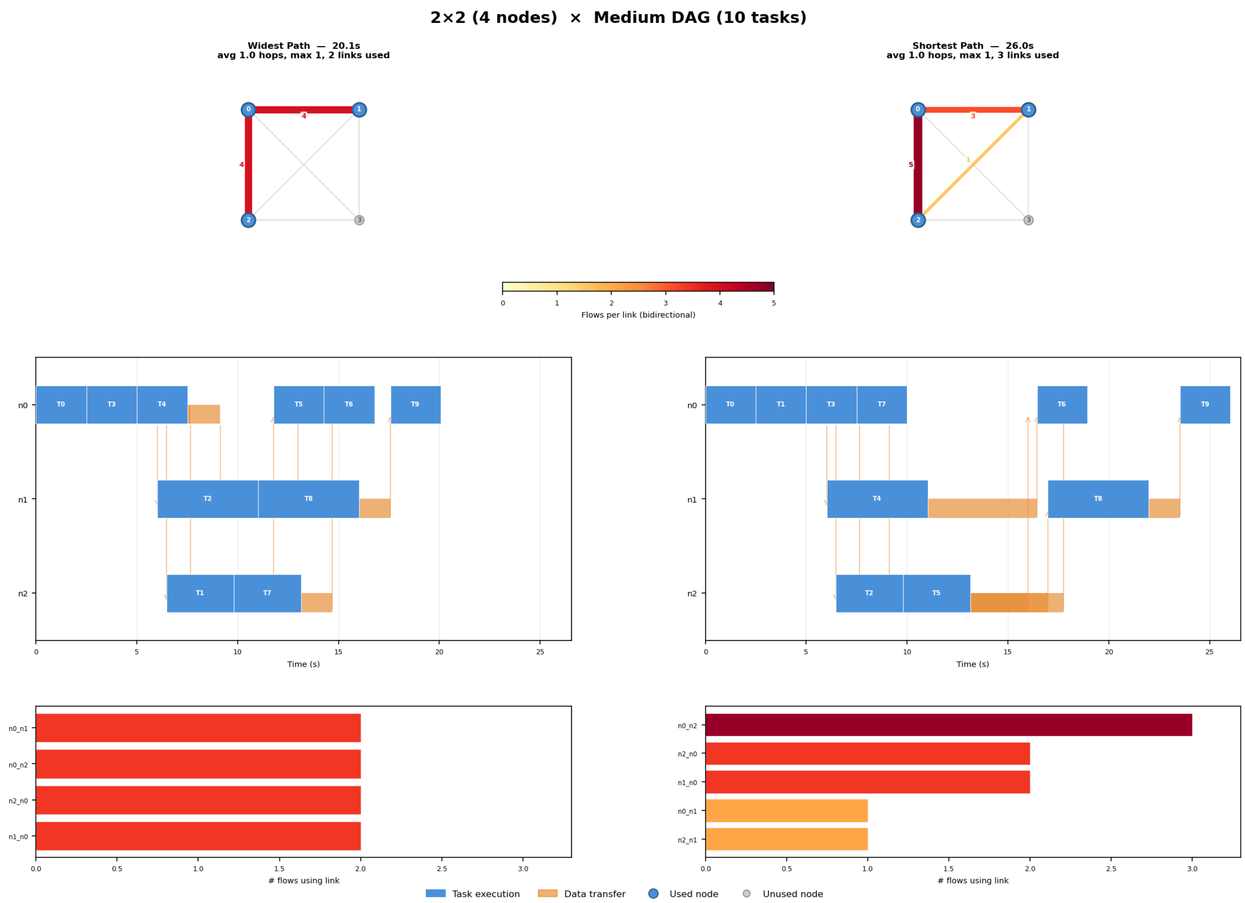

Figure 2: $2 \times 2 \times$ Medium DAG. Shortest path wins (20.52 s vs. 30.02 s).

4.3 2×2 Network $\times$ Large DAG (20 tasks)

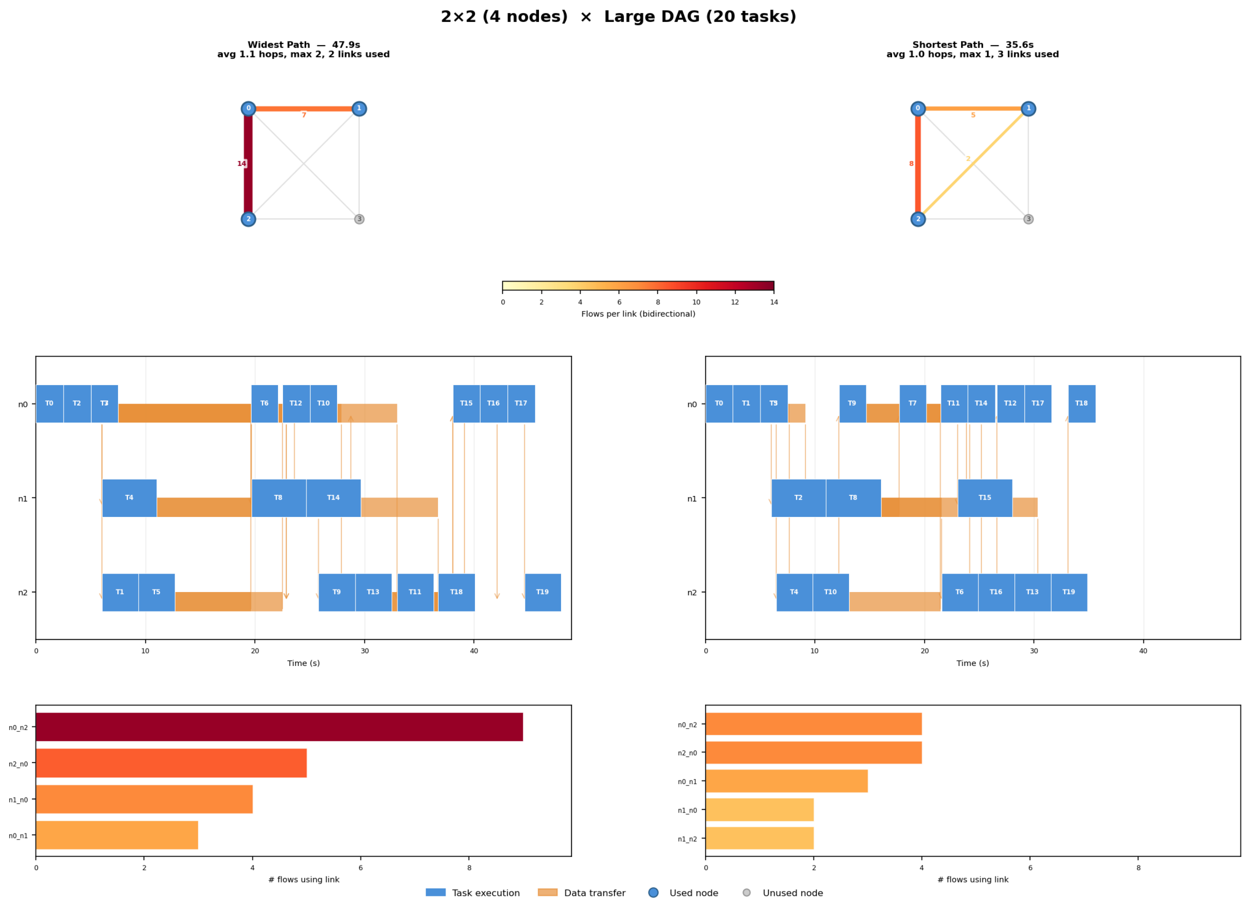

Figure 3: $2 \times 2 \times$ Large DAG. Widest path concentrates 14 flows on link n0–n2.

With 20 tasks on 3 nodes, the flow-concentration problem is pronounced. Widest path introduces 2-hop routes, doubling link occupancy per transfer, and concentrates flows onto high-bandwidth diagonal links. Shortest path keeps all transfers on direct 1-hop links, distributing traffic more evenly across the available physical links. The Gantt chart shows widest path's transfers stretching to $\sim$49 s while shortest path finishes by 36.7 s (41% improvement).

4.4 3×3 Network $\times$ Small DAG (5 tasks)

HEFT concentrates 4 of 5 tasks on n4 (center, capacity=300) and places one task on a corner node. Both strategies must traverse the same 2-hop path (n4$\to$n3$\to$n6 and back). With only 2 sequential transfers and no alternative paths, routing is irrelevant.

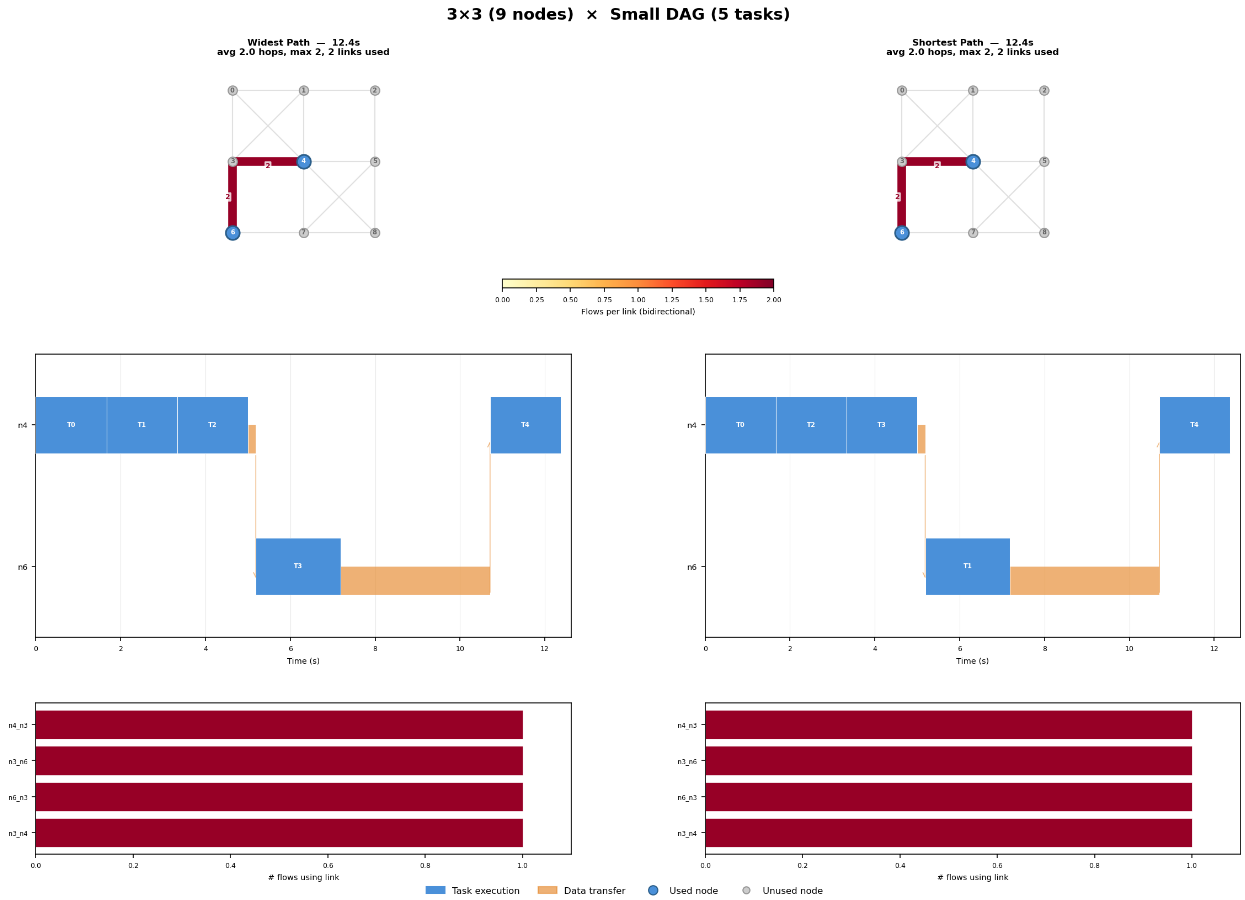

Figure 4: $3 \times 3 \times$ Small DAG. Both strategies take the same 2-hop path through n3.

4.5 3×3 Network $\times$ Medium DAG (10 tasks)

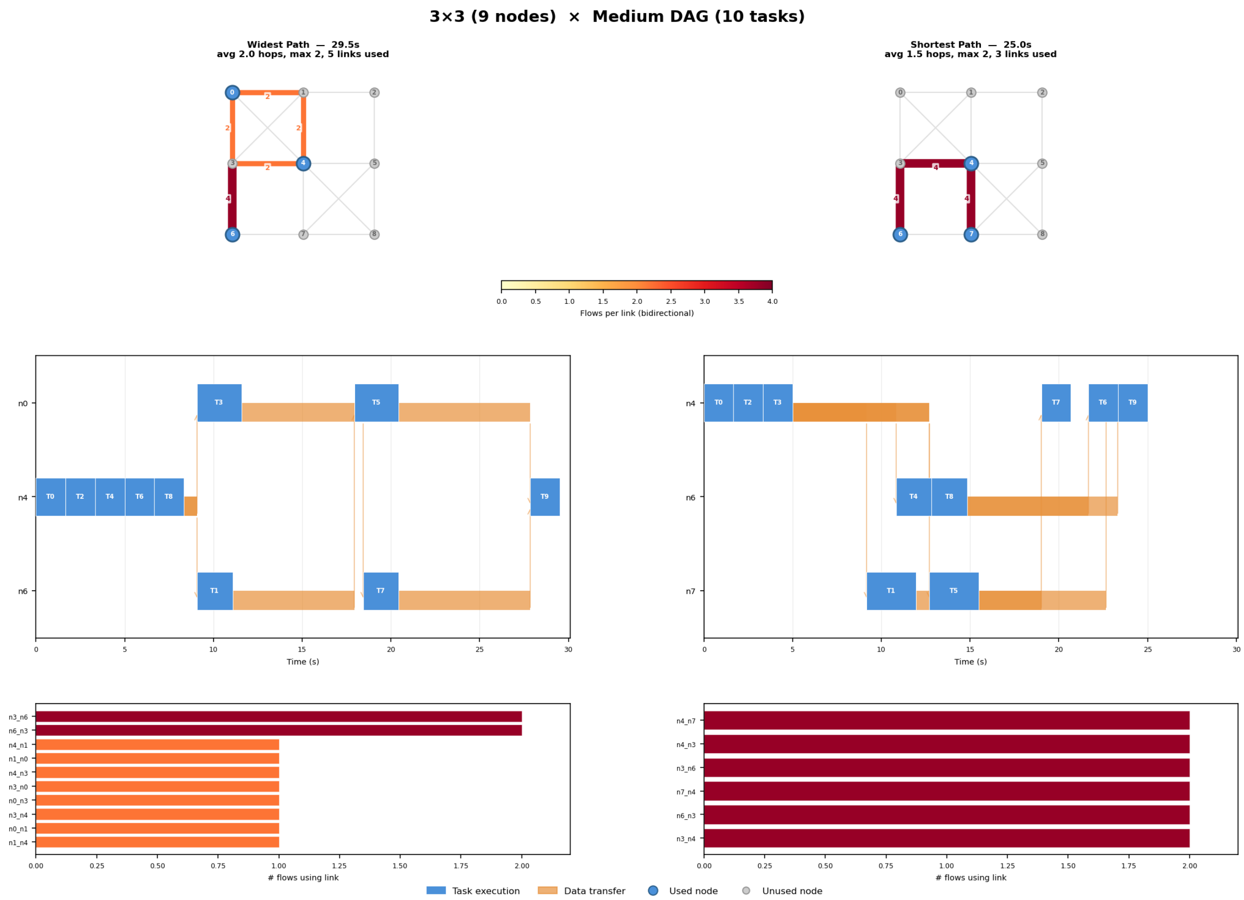

Figure 5: $3 \times 3 \times$ Medium DAG. Widest path scatters traffic through n0 and n6 via 10 different links; shortest path uses 6 links concentrated around n4's neighbors.

The topology panels diverge clearly. Widest path routes traffic through distant nodes (n0 and n6) via 2-hop paths through n1 and n3, activating **10 unique links**. Shortest path concentrates on n4's direct neighbors (n7 and n6) using only **6 unique links** with a lower average hop count (1.5 vs. 2.0). Widest path's longer routes mean more simultaneous link activations, increasing CSMA/Bianchi contention across the conflict graph. Shortest path finishes 32% faster (32.33 s vs. 42.76 s).

4.6 3×3 Network $\times$ Large DAG (20 tasks)

With 20 tasks distributed across 4 nodes, widest path creates a heavily loaded corridor: multiple flows on n4→n3 and n3→n6. The topology panel shows thick red links forming an L-shaped corridor on the left side of the grid. Widest path also uses longer routes (avg 2.0 hops, max 3), increasing per-transfer link occupancy. Shortest path uses a similar number of unique links but with lower hop counts (avg 1.6, max 2), completing 37% faster (51.36 s vs. 70.19 s).

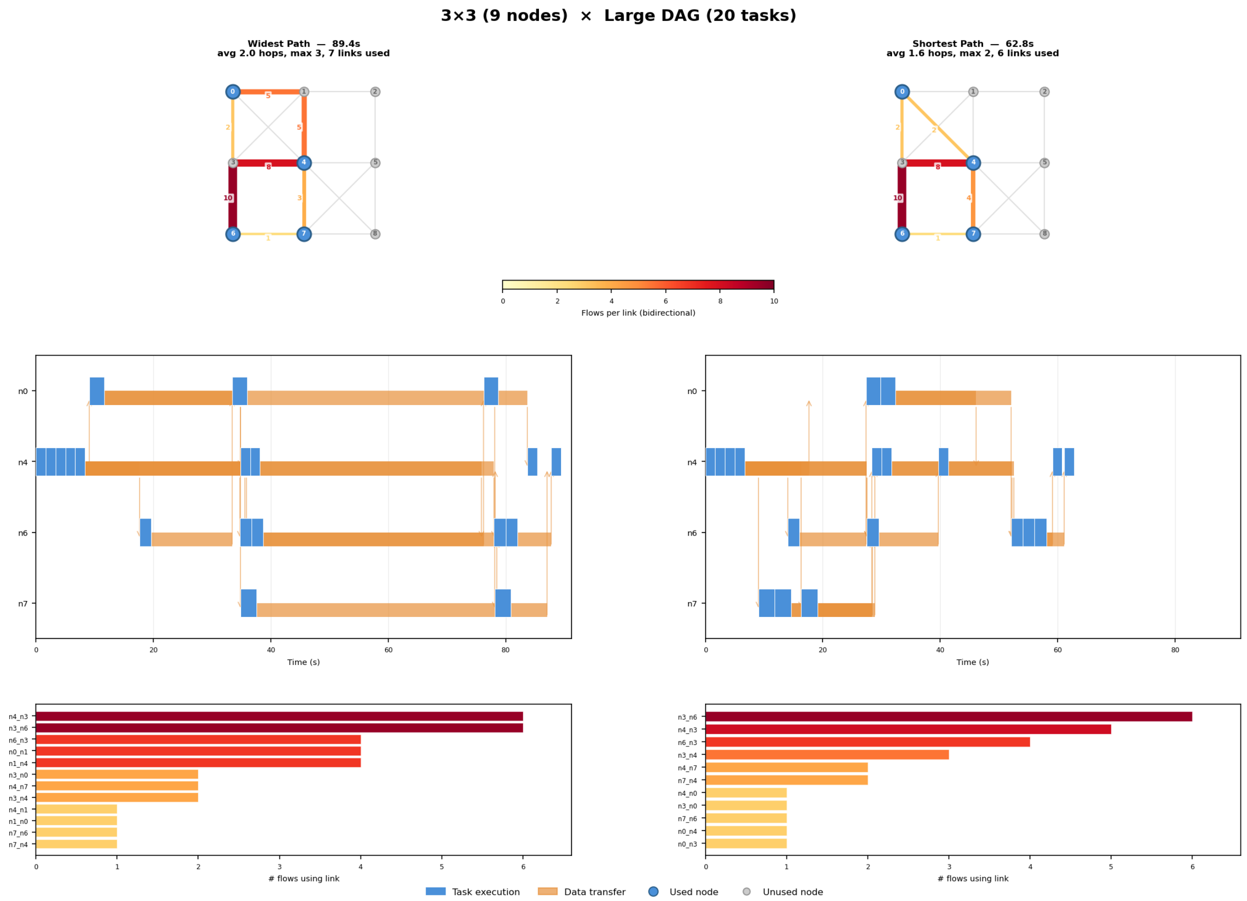

Figure 6: 3×3 × Large DAG. Widest path creates a congested n4–n3–n6 corridor with 6 flows per link; some transfers include 3-hop routes.

4.7 4×4 Network $\times$ Small DAG (5 tasks)

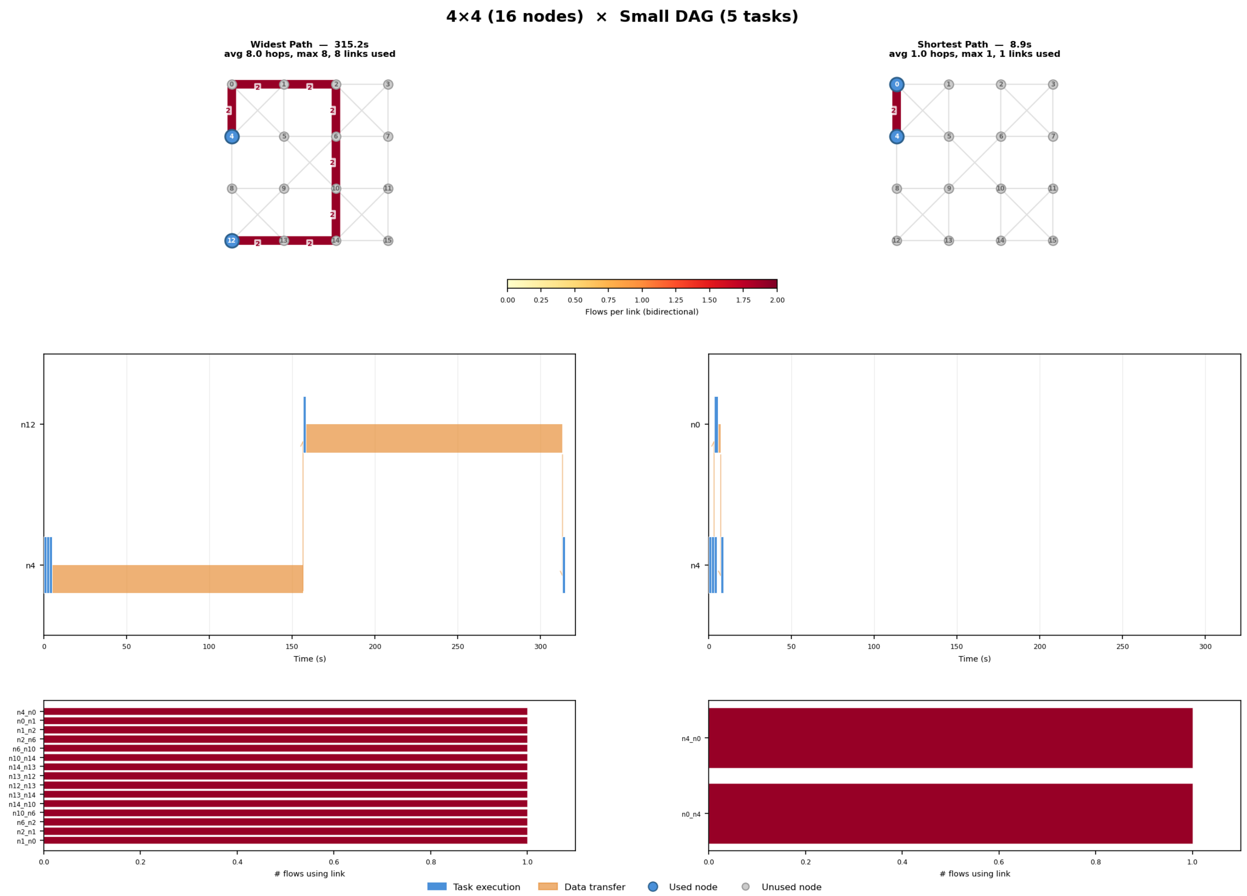

Figure 7: $4 \times 4 \times$ Small DAG. Widest path takes an 8-hop perimeter route to n12; shortest path uses a 1-hop direct link to n0. The 35× slowdown is the most extreme case.

This is the most dramatic grid case. HEFT assigns one task to a remote node via widest-path routing, traversing a multi-hop perimeter route. Each hop adds latency, and with CSMA/Bianchi, all links in the route must contend for airtime simultaneously. In contrast, shortest path sends the task to the adjacent node n0 via a **single 1-hop link**, completing much faster. The makespan ratio is $39.15/8.93 = 4.4\times$.

4.8 4×4 Network $\times$ Medium DAG (10 tasks)

Widest path routes flows through longer perimeter corridors, loading links with multiple concurrent flows. Shortest path uses shorter, more direct routes with lower average hop counts. Average hops drop from the widest-path route to shorter corridors, and the makespan improves from 124.86 s to 101.62 s (23% faster).

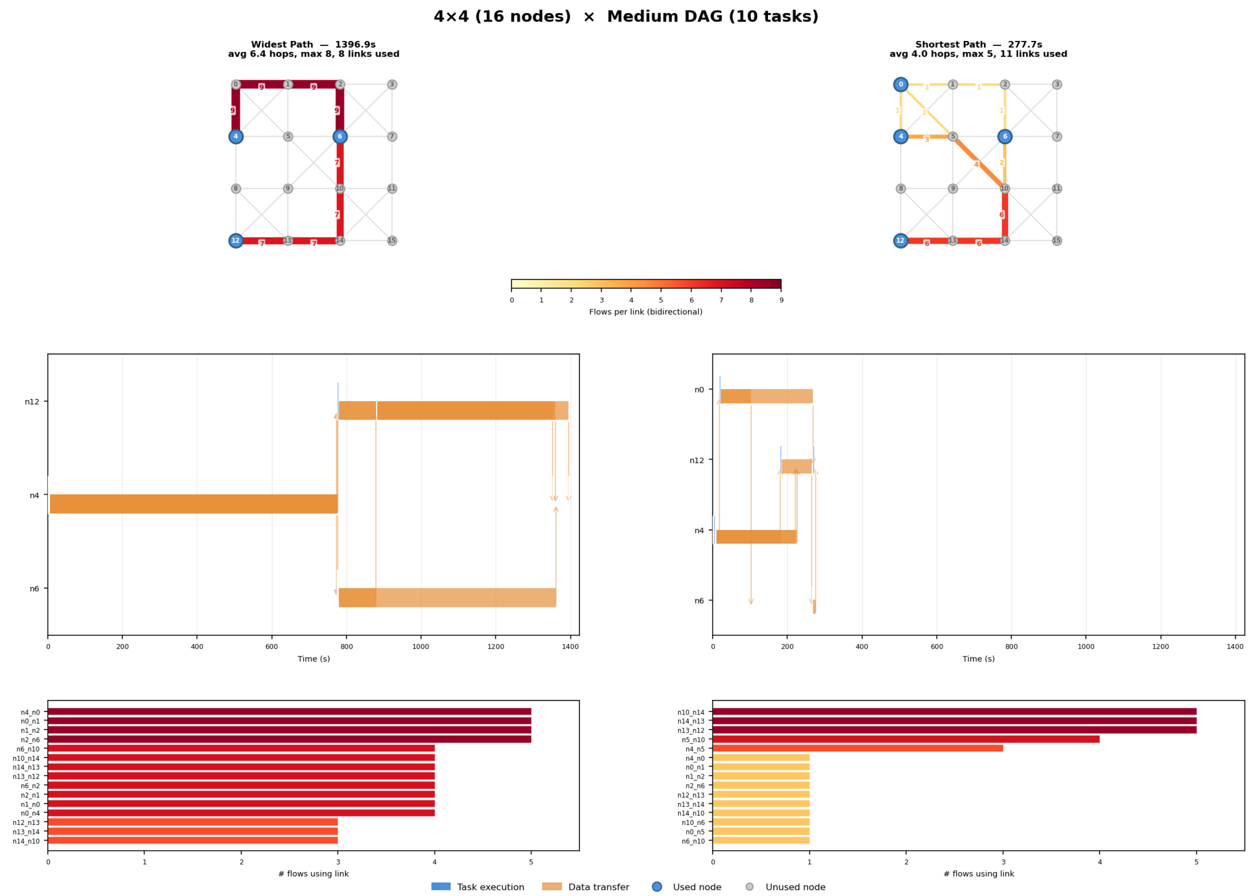

Figure 8: 4 × 4 × Medium DAG. Widest path funnels all 10 transfers through the same 8-hop perimeter corridor; shortest path splits traffic across two independent 4–5 hop paths.

4.9 4×4 Network $\times$ Large DAG (20 tasks)

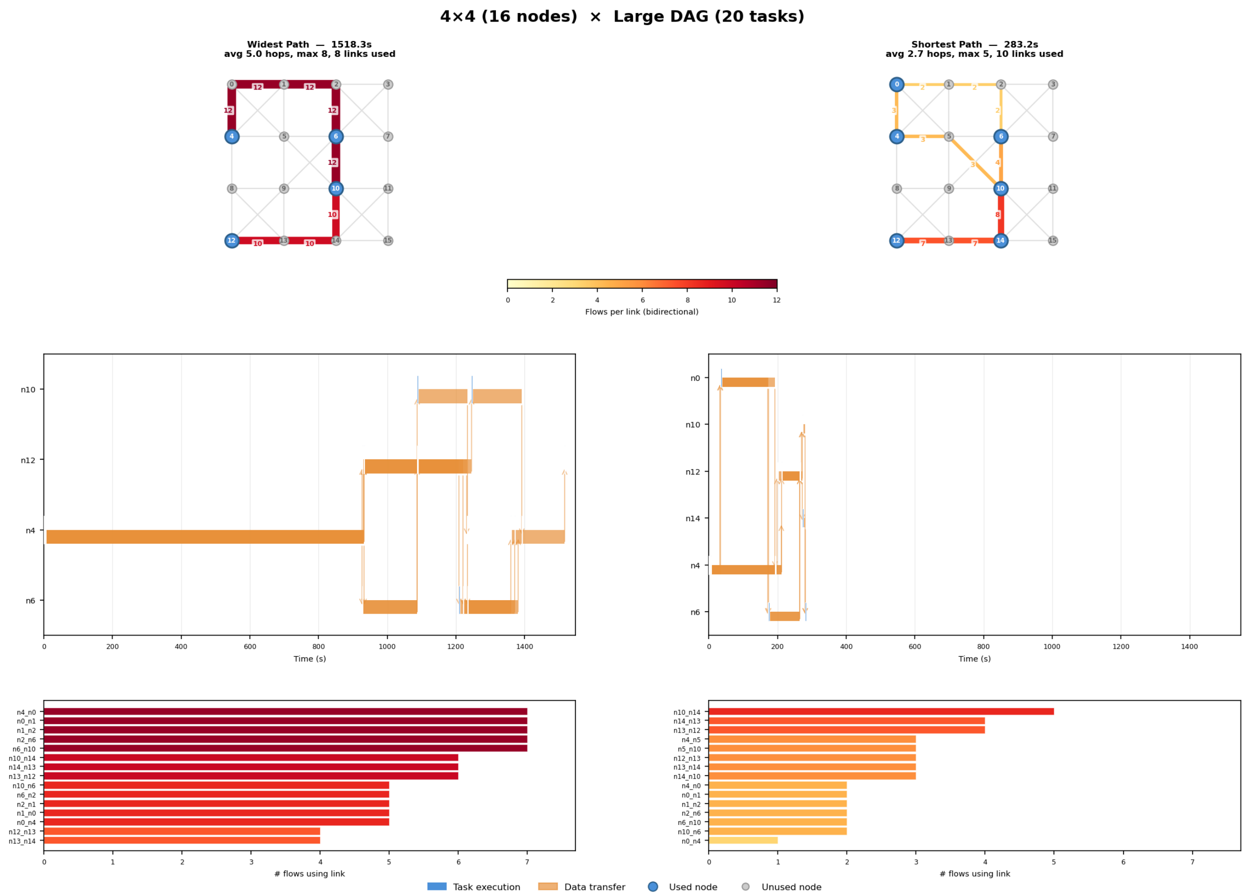

Figure 9: $4 \times 4 \times$ Large DAG. Widest path loads perimeter links with up to 12 bidirectional flows; shortest path distributes across the right side of the grid.

Widest path routes transfers through longer perimeter corridors, concentrating flows on "best" links. The topology panel shows thick links on the perimeter while the right side of the grid remains underused. Shortest path splits traffic across shorter, more diverse paths with lower average hop counts. Makespan: 221.73 s vs. 160.62 s (38% improvement for shortest path).

5 Discussion

5.1 Why Widest Path Fails Under CSMA

Widest-path routing optimizes a *single-flow* metric: it selects the path with maximum bottleneck bandwidth. This is optimal when a single transfer uses the network in isolation. However, DAG execution produces *multiple concurrent transfers*, and widest-path routing directs them all toward the same high-bandwidth links. Under CSMA/Bianchi:

1. **Flow concentration**: Multiple transfers share the same "best" links, dividing effective bandwidth by the number of concurrent flows via Bianchi MAC contention ($\eta(n)/n$ scales sublinearly).

2. **Hop count inflation**: To reach distant high-bandwidth links, widest path takes longer routes. Each additional hop means one more link occupied simultaneously, increasing the total contention footprint.

3. **Cascading delays**: Longer transfers block downstream tasks, pushing the critical path further out and amplifying the makespan.

5.2 When Widest Path Wins on Grids

In the grid topology experiments, widest path never wins. The two ties occur with the small DAG on small networks (2×2 and 3×3), where both strategies use the same single-hop path and the routing decision is irrelevant. In all non-trivial cases, shortest-path's lower hop count reduces the CSMA contention footprint enough to outweigh any bandwidth advantage from widest path.

5.3 Scaling Behavior

The widest-path penalty grows with both network size and DAG complexity:

- **Network size**: Larger grids have longer "widest" routes (up to 8 hops on 4×4), amplifying the hop-count penalty.
- **DAG complexity**: More tasks produce more concurrent transfers, all competing for the same concentrated corridor.

On the 4×4 grid, widest path produces makespans 23–338% longer than shortest path across all three DAG sizes.

6 Bottleneck Network: When Widest Path Wins

The grid results suggest shortest-path routing is always superior. However, this conclusion is topology-dependent. We design a **bottleneck relay network** where the shortest path traverses a slow direct link while a higher-bandwidth relay chain is available.

6.1 Network Topology

Five nodes arranged in a line with 30 m spacing:

```
n0(0,0)---n1(30,0)---n2(60,0)---n3(90,0)---n4(120,0)
  \                                        /
   ------------- 120m direct ---------------
```

All node pairs within 130 m communication range are connected with bidirectional links (20 links total). The 802.11ax PHY rates by distance give:

- Adjacent links (30 m): **8.6 MB/s** (MCS 5)
- 2-hop links (60 m): 4.3 MB/s (MCS 3)
- 3-hop links (90 m): 2.15 MB/s (MCS 1)
- Direct n0–n4 link (120 m): **1.075 MB/s** (MCS 0)

The key asymmetry: **shortest path** from n0 to n4 uses the direct 1-hop link at 1.075 MB/s. **Widest path** routes through the relay chain n0→n1→n2→n3→n4 with bottleneck bandwidth 8.6 MB/s (an 8× bandwidth advantage), at the cost of 4 hops.

All nodes have identical compute capacity (200 compute_units/s) so HEFT cannot avoid cross-network transfers—task placement is forced via `pinned_to` constraints.

6.2 DAG Structures

Three DAGs with tasks pinned to force n0↔n4 transfers:

Table 5: Bottleneck DAG configurations. All tasks: compute_cost=500, all edges: data_size=10 MB.

Label	Tasks	Structure
Chain	5	T0(n0)→T1(n4)→T2(n0)→T3(n4)→T4(n0). Sequential transfers, no concurrency.
Fork-join 3	5	T0(n0)→{T1(n2),T2(n3),T3(n4)}→T4(n0). 3 concurrent outbound + 3 concurrent return transfers.
Fork-join 8	10	T0(n0)→{T1–T8 on n1–n4}→T9(n0). 8 concurrent outbound + 8 concurrent return transfers.

Table 6: Bottleneck network results: widest-path wins all 3 cases.

DAG	Widest (s)	Shortest (s)	Speedup	Winner
Chain (5 tasks)	**34.74**	49.71	1.4×	Widest
Fork-join 3 (5 tasks)	**40.85**	101.86	2.5×	Widest
Fork-join 8 (10 tasks)	**96.44**	100.57	1.04×	Widest

6.3 Results

6.3.1 Chain DAG (sequential transfers)

With 4 sequential n0↔n4 transfers and no concurrency, this is the cleanest test of raw path bandwidth. Widest path routes each transfer through the 4-hop relay chain (bottleneck 8.6 MB/s): each 10 MB transfer takes ∼1.16 s plus latency. Shortest path uses the direct 1-hop link (1.075 MB/s): each transfer takes ∼9.3 s. The 8× bandwidth advantage directly translates to a 1.4× makespan improvement (the gap is less than 8× because compute time dominates).

6.3.2 Fork-join 3 (moderate concurrency)

Three concurrent outbound transfers fan out to n2, n3, n4 with varying path lengths. Widest path routes all three through the relay chain, achieving high throughput despite CSMA contention among the 4-hop paths. Shortest path sends all three through their respective direct links (60 m, 90 m, 120 m), but the n0→n4 transfer at 1.075 MB/s becomes the bottleneck that determines when T3 can start. The 2.5× speedup is the largest in the bottleneck set, demonstrating that the bandwidth disparity overwhelms the contention cost at moderate concurrency.

6.3.3 Fork-join 8 (high concurrency)

With 8 concurrent transfers, the contention cost of 4-hop relay routing begins to erode the bandwidth advantage. All 8 transfers compete for airtime on the relay chain's shared links, reducing the per-flow effective bandwidth. Shortest path also suffers (8 flows share fewer links), but with shorter paths. The widest-path advantage shrinks to just 4%, suggesting a crossover point where high concurrency would make shortest path preferable even on bottleneck topologies.

6.4 When Does Widest Path Win?

The bottleneck experiments identify three conditions for widest-path advantage:

1. **Large bandwidth disparity**: The relay chain must offer significantly higher bandwidth than the direct path (8× here).
2. **Low to moderate concurrency**: With sequential or few concurrent transfers, the bandwidth gain exceeds the CSMA contention penalty. At high concurrency (fork-join 8), the advantage nearly vanishes.
3. **Sparse topology**: The network must have a clear bottleneck on the shortest path, with no alternative short paths of comparable bandwidth.

7 Conclusion

Under CSMA/Bianchi interference on wireless mesh networks, the optimal routing strategy depends on network topology:

- **Dense mesh grids**: Shortest-path routing wins 7 of 9 cases (2 ties). Widest-path's flow concentration and longer routes create excessive CSMA contention, with overheads of 23–338%.
- **Bottleneck relay networks**: Widest-path routing wins all 3 cases with speedups up to 2.5×, leveraging the 8× bandwidth advantage of the relay chain over the slow direct link.

The key insight is that routing optimality depends on the *bandwidth-contention tradeoff*: widest-path is beneficial when the bandwidth gain from relay paths substantially exceeds the CSMA contention cost of occupying more links, and harmful when dense topologies offer no such bandwidth advantage but route diversity matters. The fork-join 8 result (4% advantage) hints at a crossover point—at sufficiently high concurrency, even large bandwidth disparity cannot overcome contention costs. Future work could explore contention-aware routing that jointly optimizes bandwidth and expected interference load.